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TD 4

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1 Single Disturbance

1.1 State-feedback with disturbance cancellation

1.1.1 State-feedback

Let us start by choosing x_1 as the output y and x_2 as the output of the first block. Then

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u + \begin{bmatrix} 1 \\ 0 \end{bmatrix} d = Ax + Bu + B_d d,$$
$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} x = Cx.$$

It as a second-order system. To choose the poles, consider $s^2 + 2\xi\omega_0 s + \omega_0^2 = 0$. To have the 5% overshoot, we set $\xi = 0.7$. Then $\omega_0 t_r \approx 3$ and from $t_r = 2$ we obtain $\omega_0 = 1.5$. We compute `K=place(A,B,poles)`.

For the input

$$u = -Kx + \ell y^r + Nd,$$

in the steady-state we have

$$y = -C\left(A - BK\right)^{-1} B\ell y^r - C\left(A - BK\right)^{-1}\left(B_d + BN\right) d,$$

and thus

$$\ell = \frac{-1}{C\left(A - BK\right)^{-1} B}$$

and

$$N = -\left(C\left(A - BK\right)^{-1} B\right)^{-1}\left(C\left(A - BK\right)^{-1} B_d\right).$$

1.1.2 Augmented system

Actually, not x nor d are known and should be estimated. We consider d as an additional state satisfying $\dot{d} = 0$ and write the augmented systems

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{d} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & -10 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ d \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u = A_a x + B_a u,$$
$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x = C_a x.$$

1.1.3 Observer design

The observability matrix is

$$\mathcal{O} = \begin{bmatrix} C_a \\ C_a A_a \\ C_a A_a^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -10 & 0 \end{bmatrix}$$

and it is of full rank. Thus, we can estimate both x and d.

To design an observer, we have to choose 3 poles. Let it be 3 times the poles of the state-feedback and -10 times ω_0:

```
observer_poles = [3*desired_poles; -10*w0]
La = place(Aa', Ca', observer_poles)'
```

The resulting controler is

$$\begin{aligned} \dot{\hat{x}}_a &= (A_a - LC_a)\,\hat{x}_a + B_a u + Ly, \\ u &= -K\hat{x} + \ell y^r + N\hat{d} = -\begin{bmatrix} K & -N \end{bmatrix} \hat{x}_a + \ell y^r. \end{aligned}$$

1.1.4 Simulations

Simulations

1.2 State-feedback with integral action

Consider now another approach, namely

$$u = -Kx + k_i \int_0^t (y^r(\tau) - y(\tau))\, d\tau.$$

Note that the integral action can replace both ℓy^r and Nd. When does such a controller solve the problem?

We start by writing the augmented state-space model. Define the new state q given by $\dot{q} = y - y^r$. The new augmented system is

$$\begin{aligned} \dot{x}_a = \begin{bmatrix} \dot{x} \\ \dot{q} \end{bmatrix} &= \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ q \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u + \begin{bmatrix} B_d \\ 0 \end{bmatrix} d + \begin{bmatrix} 0 \\ -1 \end{bmatrix} y^r \\ &= A_a x_a + B_a u + B_{d,a} d + B_r y^r, \\ y = \begin{bmatrix} C & 0 \end{bmatrix} \begin{bmatrix} x \\ q \end{bmatrix} &= C_a x_a, \end{aligned}$$

and $u = -\begin{bmatrix} K & k_i \end{bmatrix} \begin{bmatrix} x \\ q \end{bmatrix} = -K_a x_a$.

1.2.1 Conditions

We need A_a, B_a be controllable. That is satisfied if

- the number of inputs must be not less then the number of outputs; in our case $1 \geq 1$;
- A, B must be controllable - OK;
- the rank of $\begin{bmatrix} A & B \\ C & 0 \end{bmatrix}$ should be $n+p$, where p is the number of outputs, $p = \dim(y)$.

All conditions are satisfied.

1.2.2 State Estimation

To implement the integral action, we need an observer. Let us forget about the disturbance d and estimate only the state x. The pair A, C is observable, and an observer is of the form

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - \hat{y}) = (A - LC)\hat{x} + \begin{bmatrix} B & L \end{bmatrix} \begin{bmatrix} u \\ y \end{bmatrix}.$$

The gain L can be found, e.g., via pole-placement:

```
L = place(A', C', 3*desired_poles)'
```

1.2.3 Choosing k_i and trying it in simulations

The state-feedback gain vector K was computed before. How to get k_i?

Let us first keep the same K and just try some positive values of k_i. We see that the system is OK for small k_i but for $k_i \geq 5$ the system is not stable. Thus, we have to design K and k_i simultaneously.

1.2.4 Augmented system and LQ

We have to choose K and k_i together. It can be done either by pole-placement or by LQ. We do it by LQ.

Define the LQ problem. The signal u is scalar, so we can set $R = 1$. The augmented state is of order 3, so we can write

$$Q = \begin{bmatrix} Q_x & 0 \\ 0 & Q_i \end{bmatrix},$$

where Q_x is 2×2 matrix related to the states, and Q_i is a scalar weight for the state q.

Finally, `Ka = lqr(Aa, Ba, Q, R)`.

1.2.5 Summary

Finally, for the plant

$$\dot{x} = Ax + Bu + B_d d, y \quad = Cx$$

we have constructed the controller

$$\begin{aligned}
\dot{\hat{x}} &= (A - LC)\,\hat{x} + \begin{bmatrix} B & L \end{bmatrix} \begin{bmatrix} u \\ y \end{bmatrix}, \\
\dot{q} &= y - y^r, \\
u &= -K\hat{x} - k_i q.
\end{aligned}$$

1.2.6 Simulations and convergence analysis

It should work.

At the same time, if we run simulations of the observer itself, we see that it does not converge. Let us study the estimation error $e_x := x - \hat{x}$. We obtain

$$\dot{e}_x = (A - LC)\, e_x + B_d d.$$

I.e., e_x does not converge to zero but (assuming d is constant) $e_x \to -(A - LC)^{-1} B_d d$.

For the control signal $u = -K\hat{x} - k_i q$ we have

$$u = -K\hat{x} - k_i q = -Kx - k_i q + Ke_x$$

or

$$u = -\begin{bmatrix} K & k_i \end{bmatrix} \begin{bmatrix} \hat{x} \\ q \end{bmatrix} = -K_a x_a + \begin{bmatrix} Ke_x \\ 0 \end{bmatrix},$$

and the closed-loop system becomes

$$\dot{x}_a = (A_a - B_a K_a)\, x_a + \begin{bmatrix} BK \\ 0 \end{bmatrix} e_x + \begin{bmatrix} B_d \\ 0 \end{bmatrix} d + \begin{bmatrix} 0 \\ -1 \end{bmatrix} y^r.$$

We see that e_x acts in a similar way as the disturbance d, and recalling $e_x \to -(A - LC)^{-1} B_d d$, which is constant, we see that the integral action cancels the estimation error in a way similar to how it cancels the disturbance d.

Indeed,

$$y^\infty = C_a x_a^\infty = -C_a (A_a - B_a K_a)^{-1} \left(\begin{bmatrix} BK \\ 0 \end{bmatrix} e_x + \begin{bmatrix} B_d \\ 0 \end{bmatrix} d + \begin{bmatrix} 0 \\ -1 \end{bmatrix} y^r \right)$$

and

$$C_a (A_a - B_a K_a)^{-1} = \begin{bmatrix} C & 0 \end{bmatrix} \begin{bmatrix} A - BK & Bk_i \\ C & 0 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & I \end{bmatrix}$$

for any invertible matrix $(A_a - B_a K_a)$. In its turn, this matrix is invertible since it is stable due to the choice of K and k_i.